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M1 ECONOMÉTRIE ET STATISTIQUES

Spatial Models for Environmental Risks

Focus on Air Pollution

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Abstract

This project focuses on spatial modeling applied to environmental risks, with a particular interest in air pollution. From the data provided by the Geod’Air platform, we study the spatial distribution of PM10 concentrations as well as the dependence relationships between these concentrations.

To carry out this project, we rely on several tools of spatial statistics. We use variogram analysis to characterize spatial dependence, as well as kriging methods to estimate pollution levels in areas not covered by monitoring stations. Furthermore, we implement spatial regression models (SAR and SEM) in order to better understand how the pollution observed at a given point influences that of surrounding areas.

The results clearly highlight the importance of spatial dependence. When ignored, models tend to produce less accurate predictions and a biased representation of the actual distribution of pollution. Conversely, taking into account spatial interactions improves the accuracy of predictions and better captures the spatial structures of air pollution.

Introduction

Air pollution is one of the most pressing public health challenges of our time. Among the different pollutants, fine and coarse particles, PM_{2.5} and PM₁₀, attract particular attention due to their well-documented effects on respiratory and cardiovascular health. In France, as in most European countries, urban concentrations of these particles regularly exceed WHO health guidelines. According to Santé Publique France, around 40,000 deaths per year are attributed to exposure to fine particles, representing approximately 6.7% of total annual mortality, a figure comparable to alcohol-related deaths, and nearly twelve times higher than the number of road accident fatalities.

In this context, accurately identifying the most exposed areas and predicting concentrations in sectors not covered by monitoring stations has become both a scientific and a political challenge. Standard statistical approaches constitute a first tool for air quality analysis. However, they do not fully exploit the spatial information contained in the data. It is well established that geographically close stations tend to record similar pollution levels, a phenomenon known as spatial autocorrelation. Incorporating this structure into modeling therefore represents a promising avenue for improving the understanding and prediction of pollutant concentrations.

Spatial statistics offer a rigorous framework to address this limitation. Through variogram analysis and kriging, it is possible to model the spatial dependence between observations and interpolate pollution levels in unmonitored areas. Building on these foundations, spatial regression models, namely the Spatial Autoregressive Model (SAR) and the Spatial Error Model (SEM), allow explanatory variables to be incorporated while explicitly accounting for spatial structure.

This project applies these tools to PM₁₀ concentration data across France, with a dual objective: to produce more accurate predictions than non-spatial methods, and to better understand the underlying spatial dynamics of air pollution. We begin by establishing the theoretical framework, covering spatial autocorrelation, stationarity, variogram modeling and kriging, before presenting the dataset and proceeding to the empirical application and analysis of results.

Chapter 1

Background and Stakes

1.1 Why Model Pollution Spatially?

Through our research, we have found that the spatial dimension of pollution is not merely a statistical convenience, it has a physical basis. Indeed, concentrations depend on emission sources, wind patterns, topography, and atmospheric conditions, all of which vary across space. For example, urban areas and major roadways generate more pollution; valleys trap it; and the wind disperses it in certain directions.

These patterns have a real-world geographic structure, and ignoring this is tantamount to ignoring information that is actually present in the data. On the other hand, air quality monitoring stations do not cover the entire territory; there are far more unmonitored sites than monitored ones. This is where spatial interpolation comes into play, particularly kriging, which allows us to estimate pollution levels between stations and generate continuous maps.

Spatial models also help us distinguish between large-scale trends and local variations. A high reading near a road intersection may reflect a purely local effect. A widespread increase across a region may reflect industrial activity located upwind. Standard models cannot distinguish between these two cases without an explicit spatial structure.

1.2 Limitations of Non-Spatial Approaches

In linear regression, it is generally assumed that the observations are independent. However, in the case of spatially distributed pollution data, this assumption is incorrect. Stations located close to one another record correlated measurements. Indeed, two nearby points are likely to be very similar; for example, two nearby towns in rural areas are likely to have the same low concentration of fine particulate matter, and the same applies to two urban cities, which are likely to have a higher concentration of fine particulate matter. Treating nearby stations as independent leads to overestimated standard errors and tests that too often reject the null hypothesis.

More concretely : in an OLS model of pollution levels, traffic density or land use may appear to explain a large portion of the variation. But if the residuals are spatially correlated and this is ignored, it is impossible to know what is actually explained by these variables and what is simply due to the fact that neighboring areas are similar. The

coefficients may be roughly correct, but the standard errors will not be, which makes the conclusions unreliable.

The second issue concerns interpolation. If there are no monitoring stations in a given area, a standard regression will provide no information about pollution in that area beyond what the explanatory variables can predict. It cannot account for the fact that nearby stations provide relevant information. Spatial methods do take this problem into account, which is why they produce better forecasts between stations than basic econometric methods.

Local phenomena represent a third challenge. A global regression provides a set of coefficients that are valid everywhere; however, local events must be taken into account. Indeed, a valley surrounded by mountains behaves differently from a flat urban area, and the same traffic density leads to different concentrations depending on local wind patterns. Non-spatial models mask this heterogeneity. Spatial models account for this, at least in part, through the structure of neighborhoods, cities, weather, etc.

None of this means OLS is useless for environmental data. It is a reasonable starting point and gives interpretable results quickly. But for prediction at unmonitored locations, for mapping uncertainty, and for valid inference, spatial methods are necessary.

1.3 Research Objectives

The objective of this project is to study spatial models applied to air pollution, both from a theoretical point of view and through a concrete application. More specifically, we pursue the following objectives:

- Understand the key concepts of spatial statistics: stationarity, spatial autocorrelation, and variogram modeling;
- Master kriging methods so that they can be applied to a concrete case, including the adjustment of the variogram, the selection of the model, and the evaluation of the quality of the predictions;
- Define and analyze spatial regression models, including SAR and SEM models;
- Apply all of these tools to PM10 data from the Geo'Air database, in order to check whether the observed spatial structure matches theoretical expectations;
- Interpret the results and draw conclusions with a critical eye, particularly on the contribution of spatial modeling compared to classical statistical approaches.

We do not seek to produce an exhaustive study or to question existing professional work, which is often more advanced. Our goal is above all to understand and correctly use the tools of spatial statistics, then to assess their relevance and effectiveness on this type of data.

Chapter 2

Theoretical Framework: Spatial Statistics

Spatial patterns are particularly present in atmospheric data, due to local conditions and transport phenomena in the atmosphere. Ignoring spatial structure can lead to inaccurate and biased predictions. It is therefore essential to use adapted statistical tools in order to explicitly take into account the spatial dependencies present in environmental data.

2.1 Key Concepts

2.1.1 Stationarity and Spatial Dependence

All spatial models are based on assumptions about the behaviour of the process in space, the main one being stationarity. The idea is that the statistical properties of the process remain unchanged within the same geographical area. Without this assumption, it becomes impossible to generalize the behavior of the process to unobserved locations.

In geostatistics, two versions of this hypothesis are generally distinguished.

Weak stationarity

A spatial process $Z(x)$ is weakly stationary (also called second-order stationary) if two conditions hold:

- The mean is constant everywhere:

$$\mathbb{E}[Z(x)] = \mu, \quad \forall x \in \mathcal{D}$$

- The covariance between two points depends only on the vector h between them, not on their absolute positions:

$$C(x_i, x_j) = C(h) = \mathbb{E}[(Z(x_i) - \mu)(Z(x_i + h) - \mu)], \quad \forall h = x_i - x_j$$

The hypothesis that we have just seen is fundamental because it allows us to transfer the observed information to locations where we do not observe the information. Without stationarity, spatial prediction methods are not theoretically justified.

In our case, air pollution, this hypothesis is only approximatively satisfied, because major trends (urbanization, weather conditions...) affect pollution levels. This is therefore mathematically correct but often unrealistic.

Intrinsic stationarity

Because weak stationarity is often too restrictive, geostatistics mostly relies on a weaker version. Instead of requiring the process to be stationary, only the increments are. A process $Z(x)$ is intrinsically stationary if:

- The mean of any increment is zero:

$$\mathbb{E}[Z(x) - Z(x + h)] = 0, \quad \forall x, h$$

- The variance of the increment depends only on h :

$$\text{Var}(Z(x) - Z(x + h)) = 2\gamma(h), \quad \forall x$$

The function $\gamma(h)$ is the variogram, and it becomes the central object. Any weakly stationary process is also intrinsically stationary, however the converse is false. This greater flexibility is important in practice: it allows working with data that show spatial trends, without the need to correct or delete them beforehand.

Isotropy and anisotropy

After the stationarity part, we also have to decide whether spatial dependence varies with direction:

Isotropy : spatial dependence depends only on the distance $\|h\|$, not on the direction. The variogram and covariance simplify to:

$$C(h) = C(\|h\|), \quad \gamma(h) = \gamma(\|h\|)$$

Anisotropy : It refers to the fact that spatial dependence varies with direction. This means that two points located at the same distance may show different levels of correlation depending on their relative orientation.

In our context, anisotropy is often caused by wind directions: pollutants disperse more strongly along the wind direction rather than across it.

Mathematically, anisotropy can be introduced through a directional covariance (or variogram) function. Instead of depending only on the distance $h = \|\mathbf{h}\|$, the covariance depends on the full vector $\mathbf{h}$:

$$C(\mathbf{h}) \neq C(\|\mathbf{h}\|)$$

To handle anisotropy, a common approach consists in transforming the spatial coordinates so that the process becomes isotropic in a new reference frame. This is done by applying a linear transformation:

$$\mathbf{h}^* = A\mathbf{h}$$

where A is a transformation matrix that takes into account the direction and the intensity of anisotropy. The transformation includes 2 specific steps:

- **Rotation** : The first step consists in rotating the coordinate system in order to align the main axis of dependence with the dominant wind direction. This rotation is defined by the matrix:

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

where θ represents the direction of the wind.

- **Scaling (stretching)** : The second step consists in rescaling the axes to take into account the difference in spatial dependence across directions. This is done using the matrix:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\alpha} \end{bmatrix}$$

where $\alpha > 1$ is the anisotropy ratio. The first axis is aligned with the wind direction, while the second axis is perpendicular to it. For example, $\alpha = 2$ means that spatial dependence extends twice as far in the wind direction as in the perpendicular direction.

Complete transformation : The full transformation matrix is obtained by combining rotation and scaling:

$$A = R(\theta) T R(-\theta)$$

where $R(\theta)$ is the rotation matrix and T is the scaling matrix. This transformation can be interpreted as follows:

- first, rotate the coordinate system to align with the wind direction,
- then, apply a scaling to take into account anisotropy,
- finally, rotate back to the original coordinate system.

2.1.2 Spatial Autocorrelation

Definition and principles

Autocorrelation corresponds to the fact that close observations tend to be more similar than those that are distant. In this context, the autocorrelation is positive when neighboring locations are similar, and negative when neighboring locations tend to differ. Tobler's First Law of Geography says :

“near things are more related than distant things”

In practice, spatial autocorrelation takes three forms:

- Positive: similar values group together. This is the most common case for pollution.
- Negative: neighboring values contrast.
- Absent: no detectable spatial structure.

For spatial methods to work properly, autocorrelation must be present, hence the importance of systematically checking it.

Global measures: Moran's I

Moran's I is the standard global test for spatial autocorrelation. It measures whether nearby observations are more similar than chance would predict:

$$I = \frac{n}{\sum_{i,j} w_{ij}} \cdot \frac{\sum_{i,j} w_{ij} (Z(x_i) - \bar{Z}) (Z(x_j) - \bar{Z})}{\sum_i (Z(x_i) - \bar{Z})^2}$$

where n is the number of observations, w_{ij} are spatial weights encoding proximity between locations i and j , and $\bar{Z}$ is the sample mean.

The weights can be defined in various ways: geographical adjacency, a distance threshold, inverse distance... The choice affects the result and should match the scale of the process. Moran's I ranges roughly from -1 to 1 :

- $I > 0$: similar values cluster together,
- $I < 0$: neighboring values tend to differ,
- $I \approx 0$: no spatial structure detected.

Significance is assessed through a permutation test, which asks whether the observed I could plausibly arise from a random arrangement of the data.

Measuring spatial autocorrelation is an important first step to know if the use of spatial models is relevant. If the Moran index is significant, it means that nearby observations are not independent, which justifies the use of spatial statistical methods.

However, the result depends heavily on the choice of the spatial weight matrix, which introduces a certain amount of subjectivity into the analysis

Local measures (LISA)

A global statistic like Moran's I calculates one global metric of an area, it can easily miss local patterns. A region of high-high clustering surrounded by spatial randomness does not produce significant global result. To counter this problem we can introduce the Local Indicators of Spatial Association (LISA) which decomposes the global measure site by site. For each location i :

$$I_i = z_i \sum_j w_{ij} z_j,$$

This formulation can be interpreted as the contribution of each observation of the global Moran's I . The sum of all I_i is proportionally the same as the global index, which allows us to decompose spatial autocorrelation into local areas

Interpretation of the local Moran's I The local Moran statistic I_i measures the degree of spatial autocorrelation around a specific observation i . Its interpretation follows the same logic as the global Moran's I , but at a local scale.

- $I_i > 0$: positive local spatial autocorrelation. This indicates that observation i is surrounded by neighboring values similar to its own.
- $I_i < 0$: negative local spatial autocorrelation. This means that observation i is surrounded by values that differ significantly from its own.
- $I_i \approx 0$: absence of local spatial structure. In this case, there is no significant spatial pattern around observation i .

To refine this interpretation, it is useful to consider the sign of the centered variable $z_i = x_i - \bar{x}$. Combining the signs of I_i and z_i allows us to identify four typical spatial configurations:

- **High–High (HH)**: $I_i > 0$ and $z_i > 0$. A high value is surrounded by high values, forming a local cluster.
- **Low–Low (LL)**: $I_i > 0$ and $z_i < 0$. A low value is surrounded by low values, forming a homogeneous low-value cluster.
- **High–Low (HL)**: $I_i < 0$ and $z_i > 0$. A high value is surrounded by low values, indicating a spatial outlier.
- **Low–High (LH)**: $I_i < 0$ and $z_i < 0$. A low value is surrounded by high values, also corresponding to a spatial outlier.

In the context of air pollution, this classification is particularly useful to identify pollution hotspots (HH), low-concentration areas (LL), and local anomalies (HL and LH), providing a detailed understanding of the spatial structure of pollutant concentrations.

Link with the variogram and spatial modeling

Beyond global indicators, it can also be useful to look at spatial dependence in another way, by considering how it changes with distance. This is basically what the variogram does. Moran's I and the variogram are related since they both deal with spatial similarity, but they don't give the same type of information. Moran's I gives a single value for the whole dataset, so it's more like a general summary. The variogram is different because it shows how things evolve depending on distance, which is often more informative.

At short distances, nearby observations tend to have similar values, so the increase of the variogram is quite slow. Then, as distance increases, this similarity becomes weaker, and the variogram starts increasing more rapidly. It's not always perfectly smooth in practice, but the general idea stays the same.

This is important because it helps when choosing a variogram model. And depending on that choice, the predictions we obtain (for example with kriging) can be more or less reliable.

Application to air pollution

Spatial autocorrelation in air quality data is something we expect to find once we understand how pollutants behave in the atmosphere. Pollutants move and undergo chemical changes in the air over time, which means that two nearby monitoring stations will naturally measure similar levels. We can observe this phenomenon even more clearly in urban areas. Indeed, two sensors in the same neighborhood will often measure similar trends. However, there can still be exceptions; for example, a busy intersection or an industrial zone can disrupt this pattern on a small scale and create fairly distinct gradients between points that are otherwise close together.

So, in spatial studies, it's important to check for autocorrelation before diving into a geostatistical analysis—it's really useful on several levels. It helps ensure we're using the right methods and not tools that assume spatial independence that doesn't exist. It also provides an initial idea of how far spatial dependence extends in the data and helps predict whether the interpolation we intend to perform will actually hold up or not.

2.1.3 Variogram and Covariance Functions

To model spatial dependence more precisely, we need a function describing how similarity between observations decreases with distance. This is essential in order to quantify the spatial structure of the data and prepare for spatial prediction.

Definition of the variogram

The variogram is the main tool for modeling spatial dependence in geostatistics. For a spatial process $Z(x)$, it is defined as:

$$\gamma(h) = \frac{1}{2} \text{Var}(Z(x) - Z(x+h))$$

where h is the spatial lag vector between two locations. The $\frac{1}{2}$ factor is a historical convention; $2\gamma(h)$ is sometimes called the variogram without the "semi-" prefix, though both terms are used interchangeably in the literature.

Intuitively, $\gamma(h)$ measures how the average squared difference between two observations grows as the distance between them increases. For small h , nearby points are similar and $\gamma(h)$ is small. As distance increases, similarity decreases and $\gamma(h)$ grows until it levels off at a plateau. The distance at which this happens is the range, beyond which pairs of observations are effectively uncorrelated.

The variogram is preferred over the covariance function in many applications because it only requires intrinsic stationarity rather than full weak stationarity, making it valid for a broader class of processes, including those with non-constant means.

Empirical variogram

In practice, the variogram is estimated from the data using:

$$\hat{\gamma}(h) = \frac{1}{2N(h)} \sum_{\substack{i,j: \\ \|x_i - x_j\| \approx h}} (Z(x_i) - Z(x_j))^2$$

where $N(h)$ is the number of pairs with separation falling in the distance bin around h .

Computing this across a range of distances gives a point cloud showing how spatial dissimilarity evolves with distance. At large lags, fewer pairs are available and the estimate becomes noisy. At short lags, the behavior matters for kriging but is hard to estimate precisely when stations are sparse. These limitations are why a smooth theoretical model is then fitted to the empirical variogram.

Theoretical variogram models

To ensure mathematical validity and allow spatial interpolation (e.g., kriging), a parametric model is fitted to the empirical variogram. A theoretical model is required because the empirical variogram does not guarantee positive definiteness, which is necessary for kriging. The models presented below are expressed in their parametric form.

- **Exponential model**

The exponential variogram increases rapidly near the origin and approaches its limit asymptotically, making it suitable for relatively rough spatial processes.

$$\gamma(h) = c_0 + c \left(1 - \exp \left(-\frac{3h}{a} \right) \right)$$

Here, h denotes the distance between two locations.

- **Gaussian model**

The Gaussian model exhibits a parabolic behavior near the origin, reflecting a high degree of smoothness in the spatial process.

$$\gamma(h) = c_0 + c \left(1 - \exp \left(-\frac{3h^2}{a^2} \right) \right)$$

The quadratic dependence in h implies stronger continuity at short distances.

- **Spherical model**

The spherical model increases linearly near the origin and reaches its maximum value at a finite distance:

$$\gamma(h) = \begin{cases} c_0 + c \left(\frac{3h}{2a} - \frac{1}{2} \left(\frac{h}{a} \right)^3 \right), & h \leq a \\ c_0 + c, & h > a \end{cases}$$

This piecewise definition reflects a change of behavior at distance a .

- **Matérn model**

The Matérn model is a flexible family controlled by a smoothness parameter $\nu > 0$, which governs the regularity of the process:

$$\gamma(h) = c_0 + c \left(1 - \frac{1}{2^{\nu-1} \Gamma(\nu)} \left(\frac{h}{a} \right)^{\nu} K_{\nu} \left(\frac{h}{a} \right) \right)$$

Here, $\Gamma(\nu)$ denotes the Gamma function and K_{ν} the modified Bessel function of the second kind.

It generalizes several classical models:

- Exponential model when $\nu = 0.5$,
- Gaussian model when $\nu \rightarrow \infty$.

Model selection

The choice of variogram model depends on:

- the shape of the empirical variogram,
- the behavior near the origin,
- and the physical properties of the underlying process.

For instance:

- a parabolic behavior near the origin suggests a Gaussian model,
- a linear behavior suggests spherical or exponential models.

A poor choice of model can lead to inaccurate kriging predictions and unreliable uncertainty estimates.

Nugget, sill, and range

These parameters explicitly appear in the parametric formulation of the variogram models. Three parameters characterize the theoretical variogram:

- Nugget (c_0): the value at zero distance. Theoretically, $\gamma(0) = 0$, since two collocated points should be identical. However, in practice, a discontinuity at the origin is often observed, leading to $\gamma(0) = c_0$. A positive nugget reflects either measurement error or variability at spatial scales finer than the distance between stations. In air pollution data, both contribute: instruments are imperfect, and local emission sources create genuine variation at sub-station scale.
- Sill ($c_0 + c$): the plateau the variogram converges to. It equals the total variance of the process. Beyond the range, all pairs of observations are at the sill, meaning they carry no information about each other.
- Range (a): a parameter controlling the scale of spatial dependence. For some models (e.g., spherical), it corresponds to the distance at which the variogram reaches the sill, while for others (e.g., exponential), it represents a practical range beyond which observations are effectively uncorrelated.

Getting accurate estimates of these three parameters is necessary for kriging to work well. Poor estimates of the nugget or range translate directly into poor predictions and misleading uncertainty intervals.

Relationship between the variogram and the covariance function

Under weak stationarity, the variogram and the covariance function describe the same spatial dependence structure. Starting from the definition of the variogram,

$$\gamma(h) = \frac{1}{2} \text{Var}(Z(x) - Z(x + h)),$$

we expand the variance:

$$\text{Var}(Z(x) - Z(x + h)) = \text{Var}(Z(x)) + \text{Var}(Z(x + h)) - 2 \text{Cov}(Z(x), Z(x + h)).$$

Under weak stationarity, the variance is constant over space, so

$$\text{Var}(Z(x)) = \text{Var}(Z(x + h)) = C(0),$$

and the covariance between two locations depends only on the lag h :

$$\text{Cov}(Z(x), Z(x + h)) = C(h).$$

Substituting these expressions into the definition of the variogram gives

$$\gamma(h) = \frac{1}{2} (C(0) + C(0) - 2C(h)) = C(0) - C(h).$$

Thus, under weak stationarity,

$$\gamma(h) = C(0) - C(h).$$

This identity shows that the variogram increases as the covariance decreases with distance. When h is small, $C(h)$ is close to $C(0)$, so $\gamma(h)$ is small. As h increases and $C(h)$ decreases, $\gamma(h)$ grows toward its limiting value.

The variogram is often preferred to the covariance function in practice for two main reasons. First, it only requires intrinsic stationarity, which is weaker than full weak stationarity. Second, its empirical estimation does not require prior knowledge of the process mean, which makes it more robust in applied settings.

The choice of the variogram model has a direct impact on spatial predictions, making it a critical step in the analysis. Different models can lead to significantly different interpolation results. In practice, this choice is often empirical and may introduce additional uncertainty in the modeling process.

Once spatial dependence is characterized, it can be used for spatial prediction through kriging.

2.2 Kriging

Kriging is a spatial interpolation method based on the variogram structure of the data. As a result, it avoids counting the same information multiple times when observations are close together. It minimizes prediction variance while ensuring the absence of bias, making it the optimal linear estimator. It can thus be interpreted as the best linear unbiased estimator (BLUE) in the presence of spatial dependence.

Using this method is therefore very useful for our study on air pollution, where monitoring stations are few in number and predictions are required over continuous spatial domains. However, its performance depends heavily on the accuracy of the variogram model, which can be difficult to estimate in practice. There is primarily a metric called “Kriging variance” that measures the uncertainty of the predictions, i.e., how confident the model is in its prediction.

2.2.1 Ordinary Kriging

Ordinary kriging is the most commonly used variant. It assumes a constant but unknown mean over the study domain and estimates the value at an unobserved location x_0 as a weighted sum of the available observations:

$$\hat{Z}(x_0) = \sum_{i=1}^n \lambda_i Z(x_i)$$

The weights are chosen such that the estimator is unbiased:

$$\sum_{i=1}^n \lambda_i = 1$$

The kriging estimator is obtained by minimizing the variance of the prediction error:

$$\text{Var}(\hat{Z}(x_0) - Z(x_0))$$

Using the variogram $\gamma(h)$, this variance can be written as:

$$\text{Var}(\hat{Z}(x_0) - Z(x_0)) = 2 \sum_{i=1}^n \lambda_i \gamma(x_i - x_0) - \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j \gamma(x_i - x_j)$$

Minimizing this variance under the constraint $\sum_{i=1}^n \lambda_i = 1$ leads to the kriging system:

$$\sum_{j=1}^n \lambda_j \gamma(x_i - x_j) + \mu = \gamma(x_i - x_0), \quad i = 1, \dots, n$$

together with:

$$\sum_{i=1}^n \lambda_i = 1$$

where μ is a Lagrange multiplier introduced to enforce the unbiasedness condition.

Observations close to x_0 generally receive higher weights, while distant observations or those providing redundant information receive lower weights. The weights depend on the spatial structure described by the variogram.

2.2.2 Universal Kriging

Universal kriging extends classical kriging to situations where the mean is not constant across the entire study area but varies spatially. This is the case for environmental data, such as the concentration of fine particulate matter in the atmosphere, where large-scale factors such as urbanization, altitude, or proximity to emission sources influence the overall level of the variable.

In universal kriging, instead of assuming a constant mean, the spatial process is decomposed into two components: a deterministic trend and a residual term. The trend represents the large-scale variation and is generally modeled as a function of known explanatory variables or spatial coordinates. The residual component constitutes the remaining spatial dependence and is assumed to be intrinsically stationary with a mean of zero.

Using this method, we will jointly estimate both the trend and the spatial dependence within a single model. This allows the method to account for both large-scale structures and local spatial correlations, leading to more realistic forecasts when the assumption of a constant mean is inappropriate.

To model the spatial trend, we assume that it can be expressed as a linear combination of known functions of the coordinates or explanatory variables. This leads to representing the mean in the following form:

$$m(x) = \sum_{k=1}^p \beta_k f_k(x)$$

where $f_k(x)$ are known functions (covariates such as spatial coordinates or environmental variables) and β_k are unknown coefficients.

The kriging weights satisfy:

$$\sum_{i=1}^n \lambda_i f_k(x_i) = f_k(x_0), \quad k = 1, \dots, p$$

Typical covariates for air pollution studies may include:

- urbanization level and population density,
- traffic intensity along major roads,
- proximity to industrial emission sources,
- topography (valleys can trap pollutants, while elevated terrain enhances ventilation),
- land use patterns (residential, commercial, industrial).

Including such covariates helps capture large-scale variation and allows the residual component to represent local spatial dependence. Predictions are generally improved compared to ordinary kriging when spatial trends are present.

This approach is particularly relevant for environmental data, where large-scale trends (e.g., altitude, urbanization, or proximity to emission sources) are common and must be accounted for explicitly.

2.2.3 Kriging Variance and Uncertainty Quantification

One of the most useful properties of kriging is that it provides a prediction variance alongside each estimate:

$$\sigma_K^2(x_0) = \text{Var}(\hat{Z}(x_0) - Z(x_0))$$

This variance has several important properties.

- It depends only on the spatial configuration of observations and on the variogram model, not on the observed values themselves. As a result, uncertainty can be assessed independently of the data values, which is useful for planning monitoring networks.
- It decreases when observations are dense around the prediction location x_0 , reflecting higher confidence in the estimate. Near observed data points, the kriging variance becomes small, depending on the variogram model and particularly on the nugget effect.
- It increases in areas where observations are sparse or distant, correctly indicating where predictions are less reliable.

For environmental applications, the uncertainty map is often as important as the prediction map itself. In air quality studies, regions with high kriging variance indicate areas where the monitoring network may not adequately capture pollution levels.

This has several practical implications:

- identifying locations where additional monitoring stations would most effectively reduce uncertainty,
- helping public health authorities assess the reliability of pollution maps,
- supporting risk analysis by highlighting areas where predictions are less certain.

Beyond interpolation, spatial dependence can also be incorporated directly into regression models.

2.3 Spatial Regression Models

We have seen that kriging is an interpolation tool used to estimate unobserved points; we will now examine spatial regression models that explain how explanatory variables (traffic density, land use, population) are related to pollution, and how the spatial structure of the data influences this relationship.

When using the OLS method, we assume that the observations are independent. However, for spatial data, this is almost never the case, as two nearby points are often similar. Thus, ignoring spatial correlation leads to incorrect standard errors and unreliable inferences, even when the coefficient estimates are otherwise reasonable. In spatial modeling, there are two main classes of models: the Spatial Autoregressive Model (SAR) and the Spatial Error Model (SEM)

2.3.1 Spatial Autoregressive Model (SAR)

The Spatial Autoregressive (SAR) model introduces spatial dependence directly into the response variable through a spatial lag term. The model is written as:

$$Y = \rho WY + X\beta + \varepsilon$$

where:

- $Y \in \mathbb{R}^n$ is the vector of observed pollution levels,
- W is the $n \times n$ spatial weights matrix encoding the neighborhood structure,
- $\rho \in (-1, 1)$ is the spatial autoregressive parameter,
- X is the matrix of explanatory variables with coefficient vector β ,
- $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ is the error term.

The spatial weighting matrix W is used to detect whether there is a neighborhood relationship between our various observations. We can define the matrix using several criteria: geographic contiguity (units sharing a border), a distance threshold, and inverse distance. When the matrix is normalized by row, the spatial dependency WY can be interpreted as a weighted average of neighboring values.

We thus obtain:

$$(I - \rho W)Y = X\beta + \varepsilon$$

This formulation highlights that each observation is linked not only to its direct neighbors, but also indirectly to the entire spatial system through the matrix $(I - \rho W)$. Spatial dependence therefore propagates across multiple orders of neighbors, making the SAR model particularly suitable for diffusion-type phenomena. The parameter ρ controls the nature of spatial interaction:

- $\rho > 0$: similar values cluster together (positive spatial autocorrelation),
- $\rho < 0$: neighboring values tend to differ (negative spatial autocorrelation),

- $\rho = 0$: the model reduces to a standard linear regression (OLS).

Ordinary Least Squares (OLS) estimation is not appropriate in this context because the spatially lagged variable WY is endogenous, being correlated with the error term. As a result, the parameters are typically estimated using Maximum Likelihood or the Generalized Method of Moments.

In the context of air pollution, the SAR model has a clear physical interpretation. Pollutants emitted in one location are transported to neighboring areas through atmospheric processes such as wind and diffusion. The spatial lag term WY captures these spillover effects, allowing the model to represent how local pollution levels are influenced by surrounding regions.

2.3.2 Spatial Error Model (SEM)

The SEM places spatial dependence in the error term rather than in the response:

$$Y = X\beta + u, \quad u = \lambda W u + \varepsilon$$

where u is the spatially autocorrelated error, λ is the spatial error parameter, and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$.

The reduced form with covariance is:

$$\text{Cov}(Y) = \sigma^2 [(I - \lambda W)^T (I - \lambda W)]^{-1}$$

This model is appropriate when spatial dependence comes from omitted variables that are spatially structured, rather than from a genuine interaction process. In pollution studies, these might include:

- local meteorological conditions varying smoothly across space (wind speed, boundary layer height),
- micro-climate effects from vegetation, water bodies, or urban heat islands,
- unmeasured emission sources concentrated in specific areas,
- socioeconomic patterns correlated with both pollution and the included covariates.

Ignoring error autocorrelation does not bias the OLS estimates of β , but makes them inefficient and invalidates the standard errors, undermining any hypothesis test or confidence interval. SEM corrects this by modeling the residual spatial structure directly.

The key difference between SAR and SEM lies in the source of spatial dependence: SAR models direct interaction between observations, while SEM captures dependence in the error term.

Choosing between these models depends on the underlying mechanism generating spatial dependence. In the context of air pollution, spatial dependence may arise both

from physical diffusion processes and from omitted variables, making this distinction particularly important.

A limitation is that model selection can be sensitive to the specification of the spatial weights matrix.

2.3.3 Model Comparison and Selection

The choice between SAR and SEM depends mainly on what one believes to be the cause of spatial dependence in the data.

With the SAR model, we assume that pollution at one location actually influences the surrounding areas and that, in turn, those areas also influence it. This is the most natural assumption given our understanding of how pollutants move through the air.

With the SEM model, it's different. We don't assume that locations influence one another, but rather that we've omitted important variables that are themselves spatially organized. The variables we've included in the model explain a significant portion of what we observe, but what we've left out still creates a correlation in the model's errors.

In practice, we use Lagrange Multiplier (LM) tests on OLS residuals to select our model. Robust versions of these tests allow us to distinguish between the two approaches. Metrics such as AIC and BIC can also be used to compare the fitted models.

When it comes to air pollution, SAR appears to be—though no tests have yet been conducted—the best choice when the goal is to account for physical dispersion, as it incorporates pollutant transport into the model structure. SEM is more suitable when important spatial covariates are missing, for example when detailed meteorological or land-use data are unavailable. When both of these sources of dependence are present simultaneously, the more general SARMA model (spatial lags in both the response and the error term) may be necessary.

Chapter 3

Data

3.1 Sources

We used data from the *Géod’Air*, website, an open-access platform that aggregates air quality measurements from monitoring agencies. Most of the records come from *ATMO* monitoring stations, while others belong to local associations such as *Airparif* or *Air Breizh*.

The dataset is available at the following address:

`https://www.geodair.fr/donnees/consultation`

For several years, *Géod’Air* has provided historical measurements of air pollutants from monitoring stations located throughout France. These data are considered reliable for environmental analysis, public health research, and regulatory monitoring.

For our project, we are particularly interested in PM10: particles with a diameter of less than 10 micrometers. PM10 enters the respiratory system and is associated with cardiovascular and pulmonary diseases. Since it is one of the most harmful pollutants, it is also one of the most frequently measured pollutants across the network of stations, making it particularly well-suited for spatial analysis.

When exporting the data, we chose to extract the 2025 annual mean. Each observation therefore includes the station identifier, name, geographic coordinates, station type (Background, Traffic, Industrial), the 2025 annual mean PM10 value in $\mu\text{g}/\text{m}^3$, and a boolean indicator reflecting data quality and validity.

This dataset works well for spatial modeling: it provides geolocated observations across a large territory, we expect spatial autocorrelation to be present, and the uneven distribution of stations makes interpolation at unmonitored locations a meaningful task.

3.2 Data Processing

Before running our analyses, we had to go through a few processing steps to make sure the data was reliable enough to use in the models.

At first, we filtered the observations according to their validity. Each measure is associated with a boolean indicator of validity, and we kept the valid observations, to guarantee the reliability of the concentrations in PM10 used in the analyses.

In a second time, we limited the dataset to the metropolitan France, excluding overseas departments and territories. This restriction made sense, DOM-TOM are geographically isolated from the mainland. Their inclusion would have introduced spatial discontinuities, incompatible with the model assumptions of spatial regression which assume a coherent neighborhood structure between the observations.

Finally, the dataset has been completed with some explanatory variables collected with public API. The cities population was collected with the INSEE's API, the altitude of every station was obtained with the Open-Elevation's API, the weather variables (mean annual wind speed, annual rainfall, mean annual temperature) were extracted with Open-Meteo's API on the 2025 period. The number of classified installations for environmental protection (ICPE) in a 5 km radius around every station was obtained with the GéoRisques's API from BRGM. These variables, combined with the base dataset, are used for the estimation of SAR and SEM models.

Chapter 4

Numerical Application

4.1 Data Description

The dataset used in this study is made of 349 (which 298 in metropolitan France with a PM10 validate) monitoring stations of air quality, spread across the whole territory, each giving a annual mean concentration of PM10 for the 2025 year.

In order to protect public health, the European directive 2008/50/EC sets two limit values for the concentration of PM10 in ambient air since 2010 : $50 \mu\text{g}/\text{m}^3$ daily mean, with a maximum of 35 exceedance days per year allowed, and $40 \mu\text{g}/\text{m}^3$ annual mean. Under the 2024/2881/EC¹ directive, adopted in October 2024, the European Union has set new limits values applicable from 2030. This directive bring the daily mean down to $45 \mu\text{g}/\text{m}^3$ with a maximum of 18 exceedance days per year, the annual mean down to $20 \mu\text{g}/\text{m}^3$. The WHO 2021 guidelines recommand a daily mean of $45 \mu\text{g}/\text{m}^3$ with only 3 to 4 exceedance days per year allowed, and an annual mean of $15 \mu\text{g}/\text{m}^3$, a threshold that the mean concentration observed in our dataset ($15.54 \mu\text{g}/\text{m}^3$) is close to.

Statistic	Value
Number of stations	298
Mean ($\mu\text{g}/\text{m}^3$)	15.11
Standard deviation	3.34
Minimum	7.77
Q1	13.11
Median	14.57
Q3	16.88
Maximum	29.55
WHO 2021 threshold	$15 \mu\text{g}/\text{m}^3$

Table 4.1: Descriptive statistics: Annual mean PM10 concentrations, metropolitan France 2025

The mean concentration observed across all stations is $15.11 \mu\text{g}/\text{m}^3$, which is slightly above the WHO 2021 recommended threshold of $15 \mu\text{g}/\text{m}^3$. The distribution shows a positive skew, with a median of $14.57 \mu\text{g}/\text{m}^3$ below the mean, and a few extreme values

¹<https://eur-lex.europa.eu/legal-content/FR/TXT/HTML/?uri=CELEX:32024L2881>

reaching up to $29.55 \mu\text{g}/\text{m}^3$. Overall, stations recording concentrations above the WHO annual recommended threshold represent nearly half of the monitoring network of the 298 retained stations

The boxplot by influence type reveals significant differences depending on the station's context. Traffic stations show the highest concentrations and the greatest variability, with a third quartile well above the WHO threshold, which confirms the dominant role of road traffic in PM10 exposure in urban areas. Background stations on the other hand display more homogeneous concentrations, closer to the threshold, while Industrial stations show an intermediate profile.

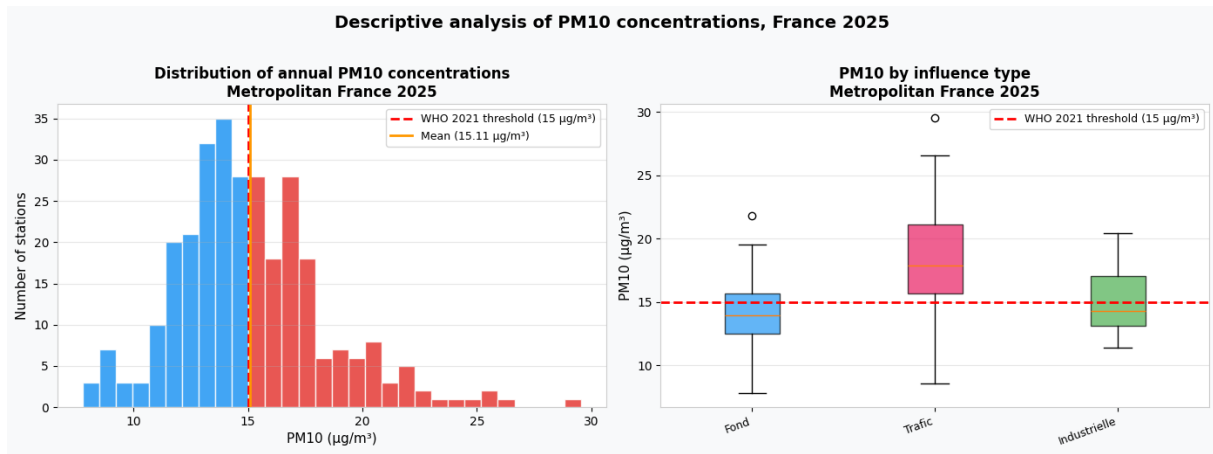


Figure 4.1: Distribution of annual PM10 concentrations and comparison by influence type, metropolitan France 2025. In red: WHO 2021 threshold ($15 \mu\text{g}/\text{m}^3$).

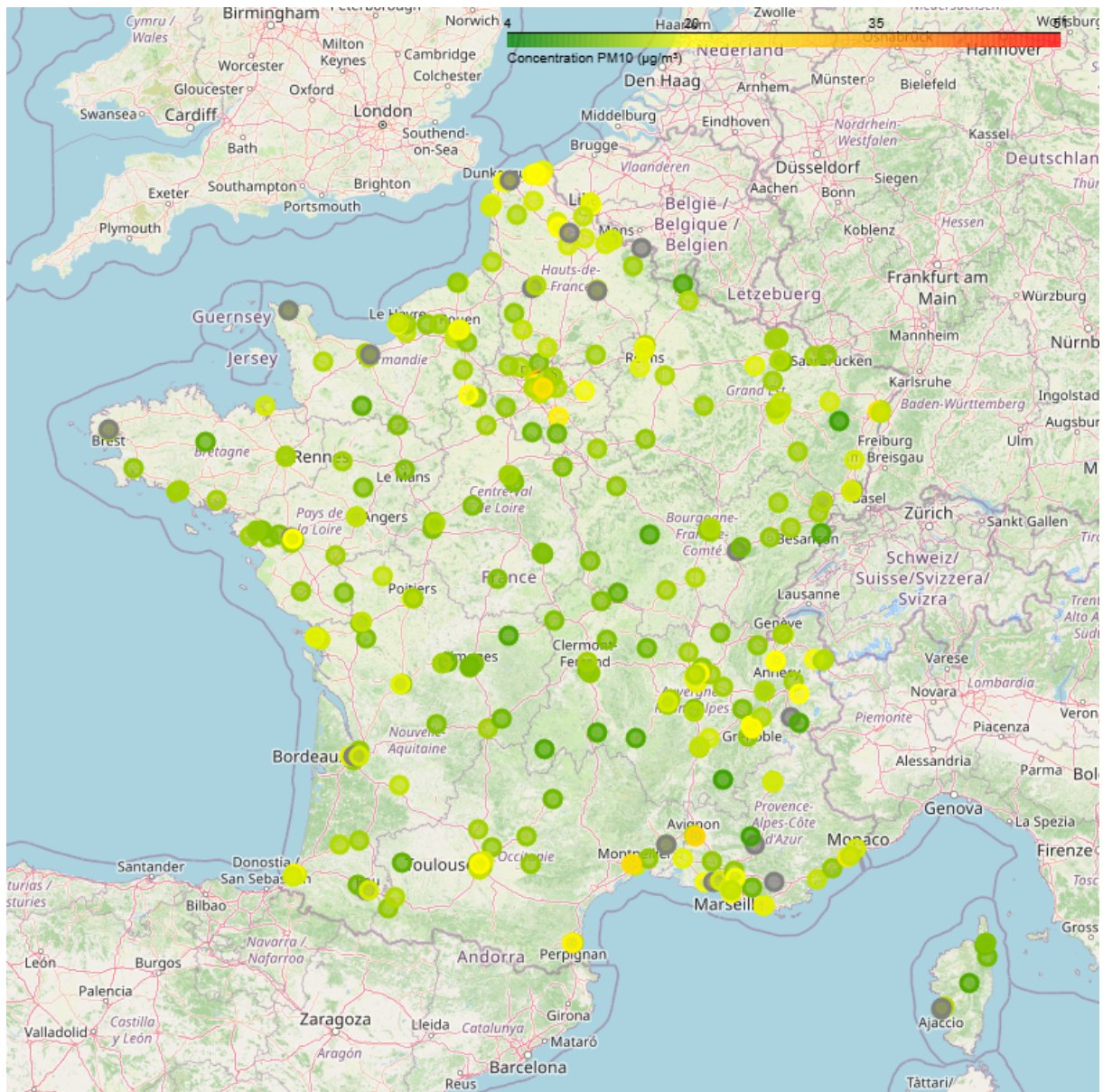


Figure 4.2: Map of PM10 concentrations by monitoring station, metropolitan France 2025

¹The interactive version is available at the following address:
https://raw.githubusercontent.com/Gookd/spatial-pollution-risk-models/main/code/carte_pm10.html

4.2 Exploratory Spatial Analysis

Before estimating the spatial models, we checked for spatial autocorrelation in the data using Moran's I, computed both on the raw PM10 concentrations and on the OLS model residuals.

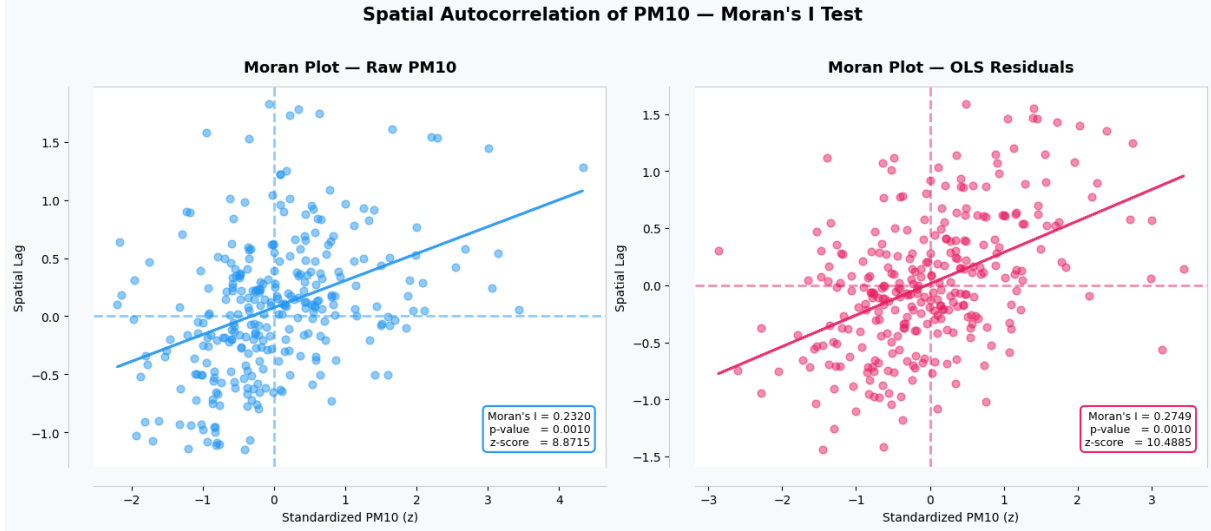


Figure 4.3: Moran plots applied to raw PM10 values and OLS residuals.

On raw PM10 values, Moran's I is 0.2320 (z-score = 8.87, p-value = 0.001), indicating a positive and significant spatial autocorrelation. Stations recording high concentrations tend to be surrounded by other stations with similarly high concentrations, and vice versa.

On the OLS residuals, Moran's I reaches 0.2749 (z-score = 10.49, p-value = 0.001), which is even higher than on the raw data. This means that even after controlling for the explanatory variables, a significant spatial structure remains in the residuals of the standard model. This result confirms that using spatial regression models, SAR and SEM, is fully justified.

4.3 Isotropy

The directional variogram plot allows us to test the isotropy assumption, meaning whether the spatial structure stays the same regardless of the direction we look at. Four directions spaced 45° apart are tested, each with an angular tolerance of $\pm 22.5^\circ$, which together cover all 180° possible directions : the 0° direction covers from -22.5° to $+22.5^\circ$, the 45° direction from 22.5° to 67.5° , the 90° direction from 67.5° to 112.5° , and the 135° direction from 112.5° to 157.5° . Since station pairs (i, j) in direction θ are symmetric to those in direction $\theta + 180$, there is no need to go beyond 180° .

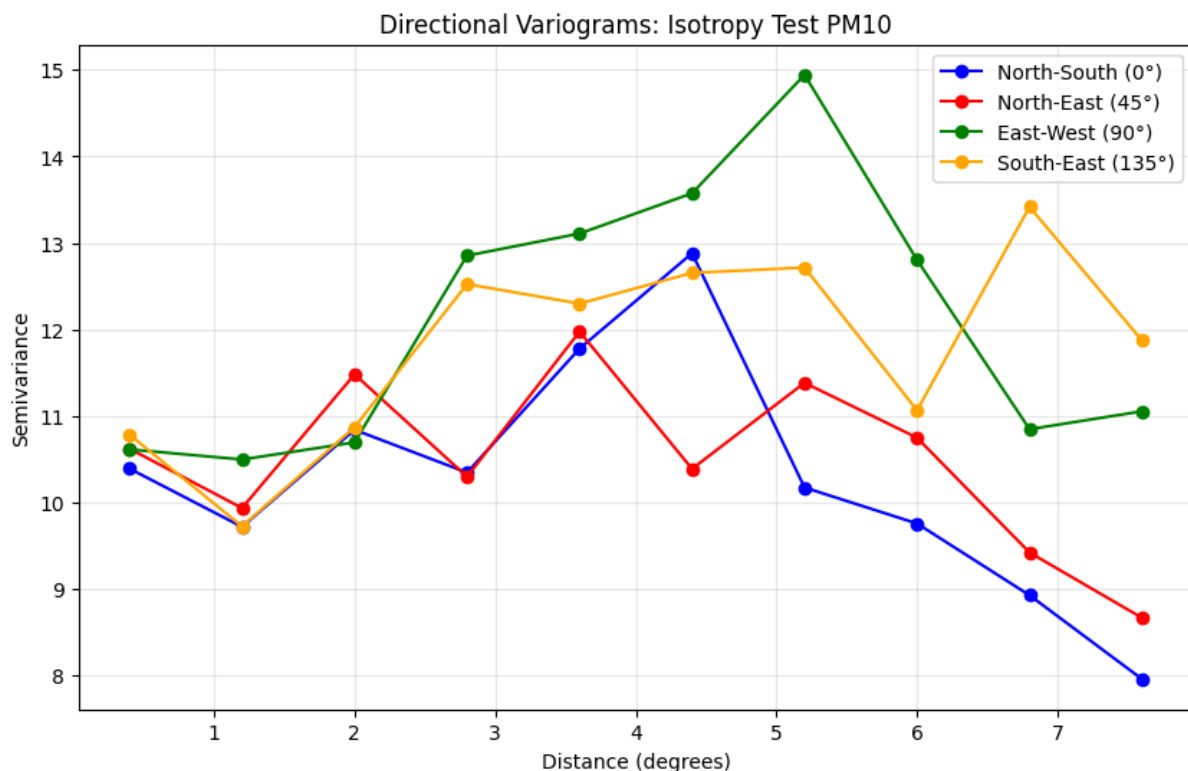


Figure 4.4: Directional variograms: isotropy test of PM10 concentrations, metropolitan France 2025.

The four tested directions show noticeably different behaviors, suggesting the presence of anisotropy in the data. In particular, the East-West direction (90°) displays the highest semivariance values, with a pronounced peak around 5 degrees of distance, while the North-South direction (0°) shows a decreasing trend at large distances, which is unusual for a stationary variogram. The North-East (45°) and South-East (135°) directions show an intermediate and more stable behavior.

These results indicate that PM10 pollution does not spread uniformly across space: variability is stronger along the East-West axis, which is consistent with the direction of the prevailing winds in metropolitan France. As a result, the kriging models were estimated taking this anisotropy into account, through the `anisotropy_scaling` and `anisotropy_angle` parameters of the `pykrige` library.

4.4 Kriging Interpolation

4.4.1 Choice of the lag numbers

The number of lags was determined by applying the standard rule according to which the product $\text{lag_size} \times \text{nlags}$ should cover approximately half the maximum distance between stations [10]. With a maximum distance of 15.14° between stations and a lag size of 0.17° corresponding to the average nearest neighbor distance, the optimal number of lags is 43, satisfying:

$$\text{lag_size} \times \text{nlags} = 0.17 \times 43 = 7.42 \approx \frac{15.14}{2} = 7.57$$

Each lag contains on average 1022 station pairs, well above the minimum threshold of 30 pairs recommended by [11], ensuring the statistical stability of the empirical variogram.

4.4.2 Choice between Ordinary Kriging and Universal Kriging

Two kriging approaches were considered: Ordinary Kriging (OK) and Universal Kriging (UK). Ordinary Kriging assumes that the mean is constant and unknown across the entire study domain. However, PM10 concentrations show a clearly non-stationary spatial structure: urban and industrial areas, mostly located in the North and North-East of France, record structurally higher concentrations than rural areas in the South-West and Brittany. Assuming a constant mean across the entire metropolitan territory would therefore be a hard assumption to justify.

Universal Kriging lifts this assumption by explicitly modeling a deterministic spatial trend (*drift*), here a linear trend in the geographic coordinates, before applying kriging on the residuals. This approach is theoretically better suited to environmental data showing a marked spatial gradient. Universal Kriging was therefore chosen for the interpolation of PM10 concentrations across metropolitan France.

4.4.3 Choice of variogram model

Three variogram models were compared: spherical, Gaussian and exponential. All three were evaluated both visually on the empirical variogram and through *leave-one-out* cross-validation. The resulting RMSE values are very close: 3.117 for the spherical, 3.143 for the Gaussian and 3.114 for the exponential. Since the visual fits were indistinguishable, the **spherical** model was selected. This choice is consistent with the physical nature of the phenomenon: air pollution has a delimited zone of influence around each source, which the spherical model captures through its finite and strict range.

Handling anisotropy

The analysis of directional variograms revealed anisotropy in the spatial structure of PM10: semivariance is higher in the East-West direction (90°) than in the North-South direction (0°), suggesting that pollution spreads more along the zonal axis. This result is consistent with the direction of the prevailing winds in metropolitan France, which predominantly blow from West to East.

To account for this, two anisotropy parameters were introduced into the Universal Kriging model. The anisotropy angle was set to `anisotropy_angle=0°`, following the `pykrige` convention which measures the angle counter-clockwise from the East-West axis, which corresponds to a main anisotropy in the East-West direction. The anisotropy scaling ratio was set to `anisotropy_scaling=1.5`, meaning that the range in the East-West direction is 1.5 times larger than in the North-South direction. This value was estimated from the directional variogram plot, by computing the ratio between the distances at which the East-West and North-South curves reach their respective sill.

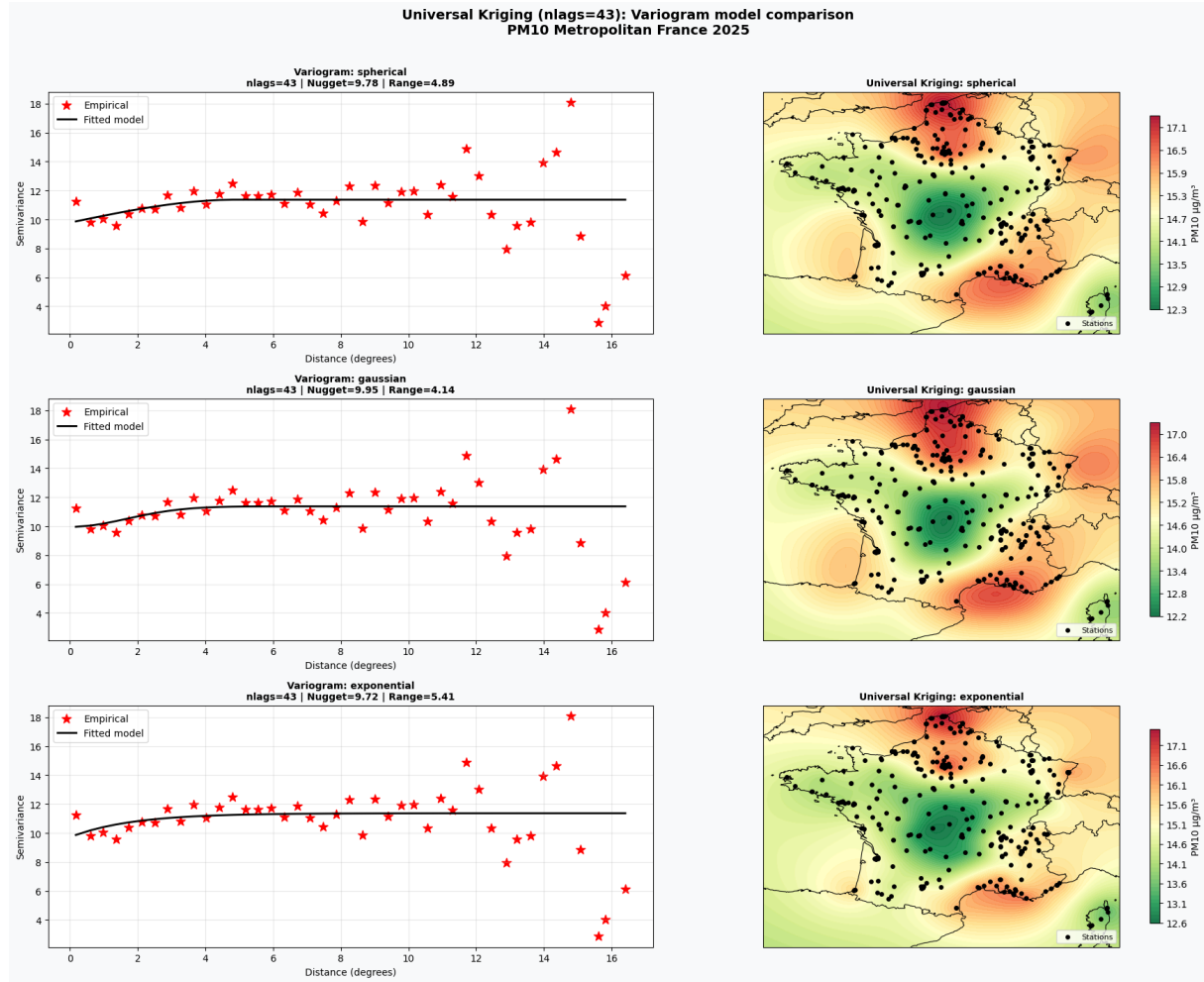


Figure 4.5: Universal Kriging, comparison of variogram models: spherical, Gaussian and exponential. Left: empirical variograms and fitted models. Right: interpolation maps of PM10 concentrations, metropolitan France 2025.

4.5 Spatial Regression

4.5.1 Model selection

Before estimating the spatial regression models, a standard OLS model was run with the `spat_diag=True` parameter in order to obtain the Lagrange Multiplier tests allowing us to choose between the SAR and SEM models. The results are presented in the following table :

Test	Value	p-value	Conclusion
LM-Lag	50.607	0.000	SAR needed
LM-Error	103.777	0.000	SEM needed
Robust LM-Lag	0.361	0.548	Not significant
Robust LM-Error	53.531	0.000	Significant

Table 4.2: Lagrange Multiplier Tests: Spatial dependence diagnostics (OLS with ICPE)

Since both basic LM tests are significant, we look at the robust versions to choose between the two models. The Robust LM-Lag is not significant ($p = 0.548$) while the Robust LM-Error is highly significant ($p < 0.001$), indicating that spatial autocorrelation mainly lies in the error term. The SEM model is therefore preferred, a result confirmed by its lower AIC compared to the SAR (1323.8 vs 1347.8).

4.5.2 Results

The three models, OLS, SAR and SEM, were estimated on 298 observations using the same explanatory variables. The following table presents the estimated coefficients and their significance.

Variable	OLS		SAR		SEM	
	Coef.	p-val	Coef.	p-val	Coef.	p-val
Constant	14.920	0.000***	8.326	0.000***	14.814	0.000***
Rural national	-3.861	0.000***	-4.031	0.000***	-3.372	0.000***
Rural near cities	-1.177	0.138	-1.295	0.073	-1.156	0.081
Rural regional	-1.439	0.052	-1.674	0.014**	-0.895	0.156
Urban	-0.605	0.127	-0.384	0.289	-0.535	0.114
Industrial	0.838	0.087	0.889	0.046**	0.861	0.048**
Traffic	3.652	0.000***	3.792	0.000***	3.548	0.000***
Altitude	-0.819	0.000***	-0.707	0.000***	-0.908	0.000***
Population	-0.048	0.841	-0.056	0.799	-0.181	0.368
Mean wind speed	0.158	0.406	0.244	0.158	0.259	0.268
Precipitation	0.356	0.046**	0.412	0.011**	0.399	0.067
Temperature	0.001	0.995	0.011	0.947	0.192	0.511
nb_icpe	0.945	0.000***	0.543	0.027**	1.098	0.000***
W_PM10 / λ	—	—	0.418	0.000***	0.598	0.000***
R²	0.503		0.567		0.497	
AIC	1380.9		1347.8		1323.8	
Log-lik.	-677.5		-659.9		-648.9	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 4.3: OLS, SAR and SEM model results. Dependent variable: PM10 ($\mu\text{g}/\text{m}^3$)

4.5.3 Interpretation

Spatial Parameter

The spatial term is highly significant in both models: $W_PM10 = 0.418$ in the SAR ($p < 0.001$) and $\lambda = 0.598$ in the SEM ($p < 0.001$), confirming the presence of strong spatial autocorrelation in the data. The higher λ in the SEM compared to the ρ in the SAR suggests that spatial dependence mainly comes from spatially structured omitted variables rather than a direct diffusion effect between stations.

Significant Variables

Traffic is the most explanatory variable across all three models, with a coefficient of $+3.55$ in the SEM ($p < 0.001$). The total impact in the SAR reaches $+6.52 \mu\text{g}/\text{m}^3$ when accounting for indirect effects, confirming the dominant role of road traffic in PM10 exposure in urban areas.

Altitude shows a significant negative effect in all three models (between -0.71 and -1.05), consistent with the physics of atmospheric dispersion: higher altitude areas benefit from better ventilation and lower pollutant accumulation.

ICPE facilities have a significant positive effect, particularly strong in the SEM ($+1.10$, $p < 0.001$), confirming that proximity to classified industrial sites contributes to higher PM10 concentrations.

Non-Significant Variables

Mean wind speed, **temperature** and **population** are not significant in the spatial models, although population is significant in the OLS. This suggests that the effect of population is captured by other variables (traffic, ICPE) once the spatial structure is taken into account. Wind and temperature, while theoretically relevant, do not seem discriminating enough at the annual and national scale to explain PM10 variations between stations.

Model Comparison

The SEM is the best performing model with an AIC of 1323.8, lower than both the SAR (1347.8) and the OLS (1393.1). This result, consistent with the robust Lagrange Multiplier tests, confirms that modeling spatial dependence in the errors is more appropriate than modeling a direct diffusion effect between neighboring stations.

4.6 Risk Assessment

4.6.1 Probability of Exceeding the WHO Threshold

From the Universal Kriging predictions and their associated variance, it is possible to estimate at each grid point the probability that PM10 concentration exceeds the WHO annual recommended threshold of $15 \mu\text{g}/\text{m}^3$. This probability is given by:

$$P(Z(x) > 15) = 1 - \Phi \left(\frac{15 - \hat{Z}(x)}{\sigma(x)} \right)$$

where $\hat{Z}(x)$ is the predicted concentration at x , $\sigma(x)$ the kriging standard deviation at x , and Φ the cumulative distribution function of the standard normal distribution.

Before applying this formula, the normality of the kriging residuals was verified. The Shapiro-Wilk test gives a statistic of 0.997 and a p-value of 0.841, failing to reject the normality assumption at the 5% level. The QQ-plot visually confirms the near-perfect alignment of the residuals along the theoretical normal line. The normality assumption is therefore fully justified for computing exceedance probabilities.

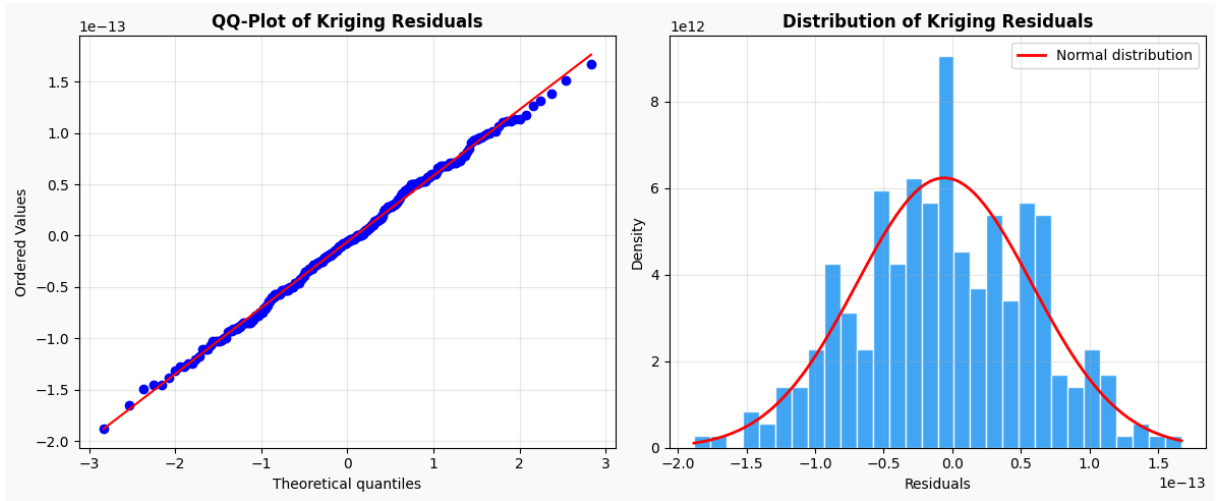


Figure 4.6: Normality check of Universal Kriging residuals: QQ-plot and histogram.

The resulting map reveals a marked spatial heterogeneity in exceedance risk across the metropolitan territory. The North and North-East of France, as well as the Mediterranean region, show the highest probabilities, reaching up to 0.75 locally. In contrast, the Centre and South-West display probabilities below 0.40, confirming their status as less exposed areas. Overall, 50.8% of the grid points show an exceedance probability above 50%, meaning that a majority of the metropolitan territory faces a significant risk of exceeding the WHO threshold.

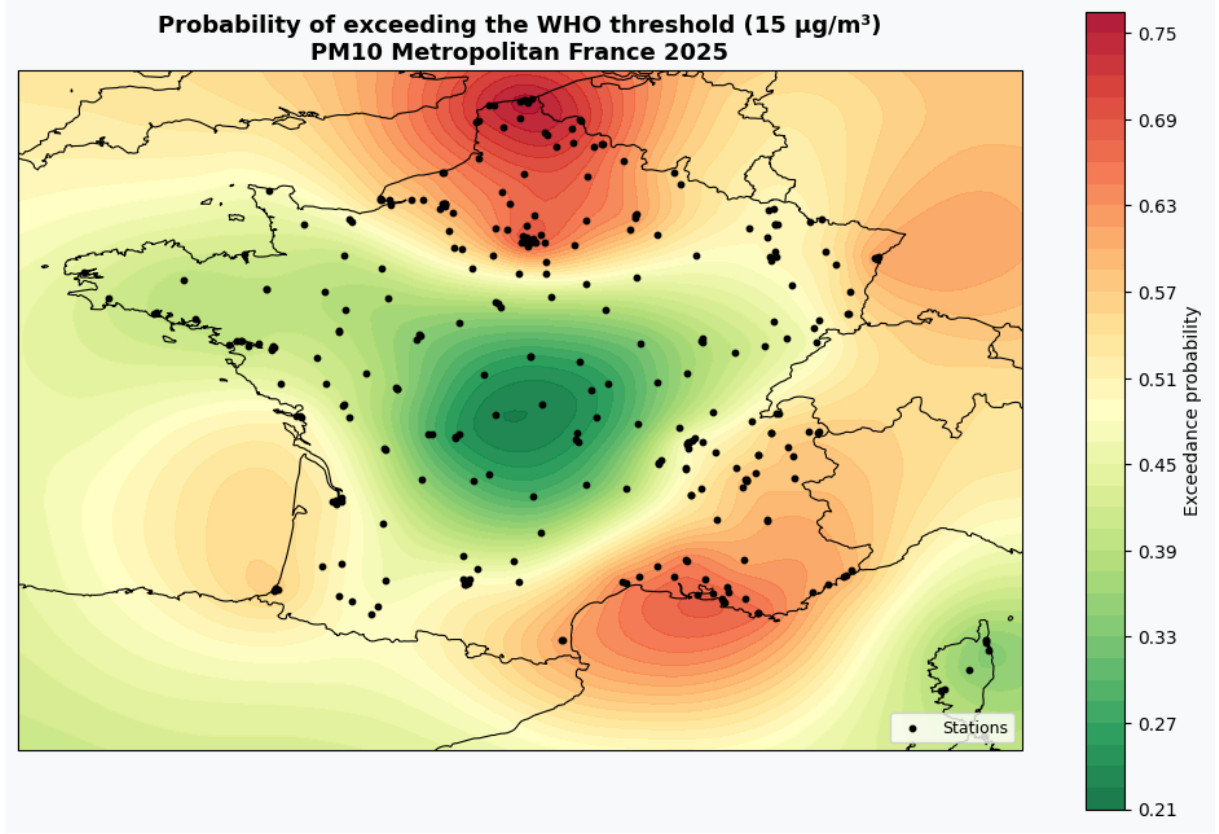


Figure 4.7: Map of exceedance probability of the WHO threshold ($15 \mu\text{g}/\text{m}^3$), PM10 Metropolitan France 2025.

4.6.2 Spatial Value at Risk

By analogy with the Value at Risk used in finance, the Spatial VaR represents the PM10 concentration level that is exceeded by only a fraction α of the grid points. Formally:

$$\text{Spatial VaR}_\alpha = \inf\{q : P(\hat{Z}(x) \leq q) \geq \alpha\}$$

The results are as follows:

Confidence level	Spatial VaR ($\mu\text{g}/\text{m}^3$)
90%	15.94
95%	16.28
99%	16.85

Table 4.4: Spatial Value at Risk of predicted PM10 concentrations, Metropolitan France 2025.

These results indicate that 90% of the territory shows predicted concentrations below $15.94 \mu\text{g}/\text{m}^3$, and that even in the most exposed areas (99th percentile), concentrations do not exceed $16.85 \mu\text{g}/\text{m}^3$ on an annual mean basis. These values remain relatively close

to the WHO threshold, highlighting the relevance of strengthened monitoring across the entire territory rather than focusing exclusively on the most exposed areas.

Chapter 5

Conclusion : Discussion of the results

The results we obtained are quite alarming and highlight a real problem of PM10 exposure in France. At this stage, half of the territory is exposed to levels above the threshold targeted by the WHO by 2030. Moreover, the most exposed areas are also the most populated, meaning the number of people affected is not proportional to the share of the territory concerned.

Looking at the regression models, areas with heavy road traffic or located near ICPE facilities (environmentally classified buildings) are the most affected by fine particle emissions. These results were rather expected and come as no surprise. However, we might have expected wind to have an impact, but according to our models it is not significant. Furthermore, rainfall has a positive effect on PM10 levels, which is counter-intuitive since rain acts as an atmospheric cleaner for fine particles. This effect may be driven by omitted variables in our model.

Our kriging model shows a RMSE of around 3, knowing that our observations typically range between 12 and 17. This means that for each prediction made by the kriging model, the error is on average $\pm 3 \mu\text{g}/\text{m}^3$ compared to the actual value. The nugget of our kriging model is also high. Both of these phenomena can be explained by the fact that geographic position alone is not sufficient to accurately infer PM10 levels in areas not covered by a monitoring station. Increasing the number of stations could help address this issue.

This type of model could be used in life actuarial science, to help estimate the risk of diseases related to fine particles based on an individual's area of residence. However, this model would obviously need to be combined with other models, given its lack of precision and the multicausality of cardiovascular, respiratory and neurological diseases.

Group Organization and Interactions with Supervisor

Our initial ideas for this research project were to work on a topic related to environmental risks, as it is a current and important subject. When Mrs. Maume-Deschamps proposed this topic, we scheduled a meeting to get a first introduction to spatial modeling. We then decided to apply this type of modeling to pollution, and more specifically to fine particles, as they affect a significant portion of the French population and present non-negligible health risks.

The group work came together quite naturally between us. We each independently started by gathering information on the subject to learn about different models and how they work. We set up two online workspaces, one on Overleaf and one on GitHub, so that everyone could have access to the various documents and to the work of the other group members. We decided to split the work among the members, in particular by separating the theoretical part from the applied part. This is where the shared workspaces became really important, as someone working on the theoretical part could check the applied part to make sure that certain concepts had not been left undefined. And someone working on the applied part could use the theoretical elements when implementing the models and analyzing the results.

AI Prompt Usage

We used AI tools (mainly Claude and ChatGPT) at several points during the project. The main uses were:

- Clarifying theoretical concepts when course material or textbooks were not clear enough,
- Getting suggestions for how to structure certain sections,
- Help with LaTeX formatting and syntax,
- Generating first-draft explanations that we then rewrote and checked against academic references.
- Help with understanding the parameters of python libraries like pykrige.

AI was not used to generate any of the results presented in this work. All outputs were produced by our own code and carefully verified.

One limitation we ran into: AI explanations of geostatistics were sometimes imprecise or mixed up terminology from different fields. We had to cross-check anything statistical against Cressie (1993) or course notes before using it. On the writing side, AI tends toward a formal vagueness that does not reflect how we actually understand things, so we rewrote most of what it generated rather than using it directly.

Overall, AI helped us move faster on understanding certain concepts by providing explanations that differed from those found in our sources.

Bibliography

- [1] Anselin, L. (1988). *Spatial Econometrics: Methods and Models*. Kluwer Academic Publishers.
- [2] Gaetan, C., & Guyon, X. (2010). *Spatial Statistics and Modeling*. Springer.
- [3] Isaaks, E. H., & Srivastava, R. M. (1989). *Applied Geostatistics*. Oxford University Press.
- [4] Cressie, N. (1993). *Statistics for Spatial Data*. Wiley.
- [5] Webster, R., & Oliver, M. A. (2007). *Geostatistics for Environmental Scientists*. Wiley.
- [6] Chilès, J.-P., & Delfiner, P. (2012). *Geostatistics: Modeling Spatial Uncertainty*. Wiley.
- [7] Pebesma, E., & Bivand, R. (2023). *Spatial Statistics for Data Science: Theory and Practice with R*. CRC Press.
- [8] Steyn, D. G., & Rao, S. T. (2010). *Air Pollution Modeling and its Application*. Springer.
- [9] Frontiers in Environmental Science (2023). *Spatial modeling of air pollution and environmental risk assessment*. Available at: <https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2023.1133178/full>
- [10] ESRI. *Choosing a lag size — ArcGIS Geostatistical Analyst*. <https://desktop.arcgis.com/en/arcmap/latest/extensions/geostatistical-analyst/choosing-a-lag-size.htm>
- [11] Pacific Northwest National Laboratory. *Kriging Variogram*. VSP Documentation. https://vsp.pnnl.gov/help/vsample/Kriging_Variogram.htm

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